# STSMN - Sensors ka kaam aur plan

Har sensor kya naapta hai, ESP32 usse kaise padhta hai, kab kya hoga (percent, points, alarm, relay, SMS, WhatsApp), aur kaise test karna hai. Saari limits **DEMO values** hain, apne hardware pe calibrate karo. Sirf low-voltage demo.

**Ek line mein:** sensor padho -> signal ko 0-100 % mein badlo -> points jodo (dhua 50 + aag 50 + baaki) -> SAFE / WARNING / EMERGENCY. EMERGENCY = buzzer + laal LED + **relay apne aap band** + SMS + app mein WhatsApp.

## 1. Poora flow: jab aag lagti hai to kya-kya hota hai

| Kab | Kya hota hai |
|---|---|
| 0 s | Flame sensor aag dekhta hai (0.5 s debounce). Flame signal 40 % se upar = asli aag. |
| turant | Risk 90+ -> **EMERGENCY**. Laal LED, buzzer, **relay band (supply kat gayi)**. App mein laal screen + notification. |
| ~1-3 s | ESP32 ka SIM800L aapke saved number pe **SMS** bhejta hai (ek baar; phir har 30 minute mein yaad dilata hai). |
| app khula ho | App WhatsApp apne aap bhejta hai (agar key saved hai). Internet na ho to har 30 s mein 6 baar koshish. |
| dhua + aag dono | Kamzor signal par bhi turant EMERGENCY (fire in progress). |
| sirf dhua | Pehle WARNING (peela). Dhua 3 s tak raha to EMERGENCY + relay band. Dhua + temperature 40 C+ ho to 2 s mein. |
| theek hone par | Signal hatne ke baad 10 s tak sab theek raha tabhi EMERGENCY se bahar aata hai. Relay phir khud 2 s stable dekh ke chalu hota hai. |

## 2. Score ka hisaab (0-100)

| Cheez | Max points | 0 % kab | 100 % kab |
|---|---|---|---|
| Dhua (MQ-2) | 50 | clean-air se 30 mV tak badhna | 450 mV badhna |
| Aag: Flame sensor | 35 | aag nahi | tez aag (AO) / DO on |
| Aag: Temperature | 15 | 40 C tak | 70 C |
| Current | 25 | 2.0 A tak | 4.5 A |
| Voltage | 15 | nominal +-15 % ke andar | +-40 % |
| Level kam | 18 | 50 % se upar | 30 % se neeche |

Kamzor signal = kam points, tez signal = poore points. SAFE: 0-54, WARNING: 55+ (45 se neeche aake SAFE), EMERGENCY: 80+ ya koi critical shart.

**Floor (score chhupa nahi sakta):** dhua/aag ka koi signal (15 %+) = kam se kam 55 (WARNING). Temperature/current/voltage apni range ke aadhe se upar = kam se kam 55 (WARNING). Koi critical = 90+.

## 3. Temperature - DS18B20 (GPIO 4)

**Kya naapta hai:** Transformer / body ka temperature.

**Kaise kaam karta hai:** Digital sensor, ek wire (OneWire). 4.7k resistor data aur 3.3V ke beech.

**ESP32 ka rule (config.h):** 40 C se points shuru, 70 C pe 100 %. 70 C+ lagatar 3 s = EMERGENCY. Sensor khula / galat reading = 'not valid' (kabhi 0 C nahi dikhata).

**Safe test:** Sensor ko haath mein pakdo (~33 C) ya gunguna paani (50 C). App mein temperature aur % badhna chahiye.

**Calibration:** Zaroorat nahi (factory calibrated). Multimeter thermometer se check kar lo.

## 4. Current - ACS712 5A (GPIO 34)

**Kya naapta hai:** Load ka current (Ampere).

**Kaise kaam karta hai:** Current badhne par output voltage badhta hai. Sensor 5V ka hai, isliye ESP32 ko 10k+20k divider se do. Idle par ~2.5 V (divider ke baad ~1.67 V).

**ESP32 ka rule (config.h):** 2.0 A se points, 4.5 A pe 100 %. 4.5 A+ lagatar 1.5 s = EMERGENCY. Idle level galat (200-3000 mV ke bahar) = sensor not valid.

**Safe test:** Chhota low-voltage load (bulb/LED strip) lagao. Pehle bina load ke 0.00 A aana chahiye.

**Calibration:** Bina load ke boot karo (zero point khud set hota hai). Multimeter se ek known load check karo.

## 5. Voltage - 0-25 V module (GPIO 35)

**Kya naapta hai:** Supply voltage.

**Kaise kaam karta hai:** Module voltage ko 5 se divide karta hai. ESP32 reading ko wapas multiply karta hai.

**ESP32 ka rule (config.h):** Nominal 3.7 V (demo battery). +-15 % tak koi points nahi, +-40 % pe 100 %. **Agar module laga nahi to config.h mein HAS_VOLTAGE_SENSOR = false karo** (warna fault maana jata hai).

**Safe test:** Battery ko multimeter se naapo aur app se milao.

**Calibration:** VOLT_CAL_FACTOR ko multimeter ki reading / app ki reading se multiply karo.

## 6. Dhua - MQ-2 (GPIO 32)

**Kya naapta hai:** Dhuaan / gas ka signal. (Ye 'kitna dhua' ka rough percent hai, ppm nahi.)

**Kaise kaam karta hai:** Heater garam hota hai (60 s warm-up, tab tak 'smoke' nahi bolta). Dhua aane par analog voltage badhta hai. Module 5V ka hai, isliye 10k+20k divider.

**ESP32 ka rule (config.h):** Clean-air baseline khud dheere-dheere set hota rehta hai. Baseline se 120 mV upar = dhua pakda gaya (60 mV neeche aane par band). 30 mV se 450 mV tak = 0 se 100 %. Dhua akela 3 s = EMERGENCY. Serial Monitor mein `mq=asli/baseline` dikhta hai.

**Safe test:** Agarbatti ya bujhi hui maachis ka dhua sensor ke paas (aag mat lagao). App mein smoke % badhta dikhna chahiye. Dhua hatao, % wapas 0 hona chahiye.

**Calibration:** Sensor ko 5 minute garam hone do. Agar bahut zyada/kam sensitive ho to MQ2_DELTA_ENTER_MV aur MQ2_FULL_DELTA_MV badlo.

## 7. Aag - Flame sensor (DO: GPIO 25, AO: GPIO 36)

**Kya naapta hai:** Aag ki infrared roshni.

**Kaise kaam karta hai:** AO pin = signal kitna tez (mukhya). DO pin = comparator ka haan/na (sirf madad). AO ko ambient baseline se compare kiya jaata hai; suraj/lamp ki dheere badalti roshni se bachne ke liye flicker, achanak badlaav aur DO bhi dekhte hain.

**ESP32 ka rule (config.h):** AO baseline se kaafi upar + (DO ya flicker ke saath achanak badlaav) 0.5 s tak = EMERGENCY. Sirf DO = EMERGENCY nahi (WARNING). AO kharab/ulta ho to DO-only fallback. Details: docs/FLAME_AO_CALIBRATION.md.

**Safe test:** Door se ek chhoti jwala (safe jagah, paani paas, adult ke saath). Suraj/lamp se EMERGENCY nahi aana chahiye.

**Calibration:** FLAME_AO_CALIBRATION.md ke steps se apni readings lo; config.h ke FLAME_* values placeholders hain. Suraj aur aag ko sensor akela alag nahi kar sakta.

## 8. Oil level (demo: sirf paani) - Water level sensor (GPIO 33)

**Kya naapta hai:** Paani ka level (asli transformer oil ke liye ye sensor nahi chalega).

**Kaise kaam karta hai:** Jitna zyada doobta hai, voltage utna badhta hai.

**ESP32 ka rule (config.h):** Level 30 % se neeche = low (10 points). 50 % se upar = theek.

**Safe test:** Sensor ko dheere paani mein dubao, % badhna chahiye.

**Calibration:** LEVEL_DRY_MV (sukha) aur LEVEL_FULL_MV (poora gila) mV note karke daalo. Laga nahi to HAS_LEVEL_SENSOR = false.

## 9. GPS - NEO-6M (GPIO 16, 17)

**Kya naapta hai:** Asli location.

**Kaise kaam karta hai:** Satellite se fix. Pehli baar 1-5 minute, **khule aasman ke neeche**. Ghar ke andar fix nahi milta.

**ESP32 ka rule (config.h):** Fix na ho to app demo/configured location dikhata hai ('LIVE GPS' nahi). 10 s koi sahi data na aaye to 'no fix'.

**Safe test:** Chhat pe ya khidki ke paas rakho. Serial Monitor mein gpsFix / lat / lng.

**Calibration:** Zaroorat nahi.

## 10. SIM800L - SMS (GPIO 13, 14)

**Kya naapta hai:** SMS alert.

**Kaise kaam karta hai:** 2G modem. **Jio SIM nahi chalega** (2G chahiye). Alag 4V 2A power + 1000 uF capacitor.

**ESP32 ka rule (config.h):** EMERGENCY mein ek SMS, do SMS ke beech kam se kam 2 minute, aur jab tak khatra rahe har 30 minute mein ek.

**Safe test:** App > Settings > Alert number daalo. Fir (low-voltage demo mein) dhua dikhao. Ya Serial Monitor mein 'SMS' lines dekho.

**Calibration:** Zaroorat nahi.

## 11. Relay, Buzzer, LED

**Kya naapta hai:** Supply kaatna, awaaz, status.

**Kaise kaam karta hai:** Relay: load COM + NO se. ESP32 band ho to bhi supply kat jaati hai. Boot par relay band, ~5 s baad sab sensor theek ho to chalu. Buzzer: EMERGENCY mein beep. LED: hara SAFE, peela WARNING, laal EMERGENCY.

**ESP32 ka rule (config.h):** RELAY_ACTIVE_LOW = true aur RELAY_OPEN_DRAIN_OFF = true (3.3V se 5V module ko band karne ke liye pin chhod diya jata hai).

**Safe test:** Boot par relay module ki LED band, ~5 s baad ON, EMERGENCY mein OFF. Ulta ho to RELAY_ACTIVE_LOW badlo.

**Calibration:** -

## 12. App mein kya dikhta hai

Sensor Data screen: har sensor ka card, chart aur percent. Dhua aur aag ke chart 0-100 % ke fixed scale mein hain: kamzor signal chhota, tez signal upar tak. Oil level card ke neeche 'Signal strength' card mein sab sensor ka percent (none / weak / medium / strong).

## 13. Seemayein (sach)

- Ye sab asli hardware par abhi test nahi hua; limits DEMO hain.
- MQ-2 ppm nahi batata, sirf relative signal.
- WhatsApp tabhi jaata hai jab app khula ho aur phone mein internet ho; SMS app ke bina bhi jaata hai.
- Real transformer par lagane se pehle licensed electrician aur proper protection zaroori hai.